«entity»
model::GridCell

- x: int
- y: int
- isPassable: boolean
- isFogOfWar: boolean

+ setPassable(passable: boolean): void
+ getCoordinates(): Point

Responsibilities
-- Represents a single cell within the
2D occupancy grid.

1..*

cells

1

«boundary»
ui::GridCanvasGUI

- canvas: Canvas
- zoomLevel: double

+ initializeCanvas(): void
+ renderGridMatrix(matrix: GridCell[][]): void

Responsibilities
-- Renders the interactive 2D matrix
map as an observer.

1

engine

1

«engine»
engine::GridEngine

- width: int
- height: int
- cells: GridCell[][]

+ initializeGrid
(width: int, height: int, preset: String)
: void
+ updateSimulationTick(): void
+ clearFogOfWar(bounds: Rectangle): void

Responsibilities
-- Manages 2D grid matrix state and
simulation tick loops.

«interface»
engine::GridObserver

+ onGridStateUpdated(): void
+ onEntityMoved(entityId: String, newPos: Point): void

Responsibilities
-- Defines subscription contract for real-time updates.

1

engine

1

«engine»
engine::ActionVerifier

- gridEngine: GridEngine

+ validateAction
(command: Command, robot: Robot)
: boolean
+ checkBatteryMargin
(robot: Robot, requiredEnergy: double)
: boolean
+ checkObstacleCollision
(destination: Point): boolean

Responsibilities
-- Central deterministic rule validation engine
intercepting execution.